A Meta-Transfer Objective for Learning to Disentangle Causal Mechanisms

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Code available on Github

Introduction

Idea 1

- What are the right representations?
- Causal variables explaining the data

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- What are the right representations?
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Idea 2

- How to modularize knowledge for easier re-use & adaptation, good transfer?
- How to disentangle the unobserved explanatory variables?

Hypotheses about how the environment changes

Main Assumptions:

- Changing one mechanism does not change the others (Peters, Janzig & Scholkopf 2017)
- Non-stationarities, changes in distribution, involve few mechanisms (e.g. the result of a single-variable intervention)

Claims

Under the hypothesis of independent mechanisms and small changes across different distributions:
smaller sample complexity to recover from a distribution change
E.g. for transfer learning, agent learning, domain adaptation, etc.

Learning a Causal Graph with two Discrete Variables

If we have the right knowledge representation, then we should get fast adaptation to the transfer distribution when starting from a model that is well trained on the training distribution

Core Idea: A "Regret" function based on the speed of adaptation.

However it is clear to us that much more work will be needed to evaluate the proposed approach in a diversity of settings and with different specific parametrizations, training objectives, environments, etc.

Let both A and B be discrete variables each taking N possible values and consider the following two parametrizations

$$P_{A\to B}(A,B) = P_{A\to B}(A)P_{A\to B}(B \mid A)$$
$$P_{B\to A}(A,B) = P_{B\to A}(B)P_{B\to A}(A \mid B)$$

This amounts to four modules: $P_{A\to B}(A)$, $P_{A\to B}(B \mid A)$, $P_{B\to A}(B)$ and $P_{B\to A}(A \mid B)$. We will train both models independently.

Maximum likelihood estimation of these parameters: normalized relative frequencies.

θ: parameters of all these models

$$\theta_{A|B}, \theta_{B|A}, \theta_B, \theta_A$$

$$\theta_i = P_{A\to B}(A = i) \qquad \theta_{j|i} = P_{A\to B}(B = j \mid A = i)$$
$$\eta_j = P_{B\to A}(B = j) \qquad \eta_{i|j} = P_{B\to A}(A = i \mid B = j).$$

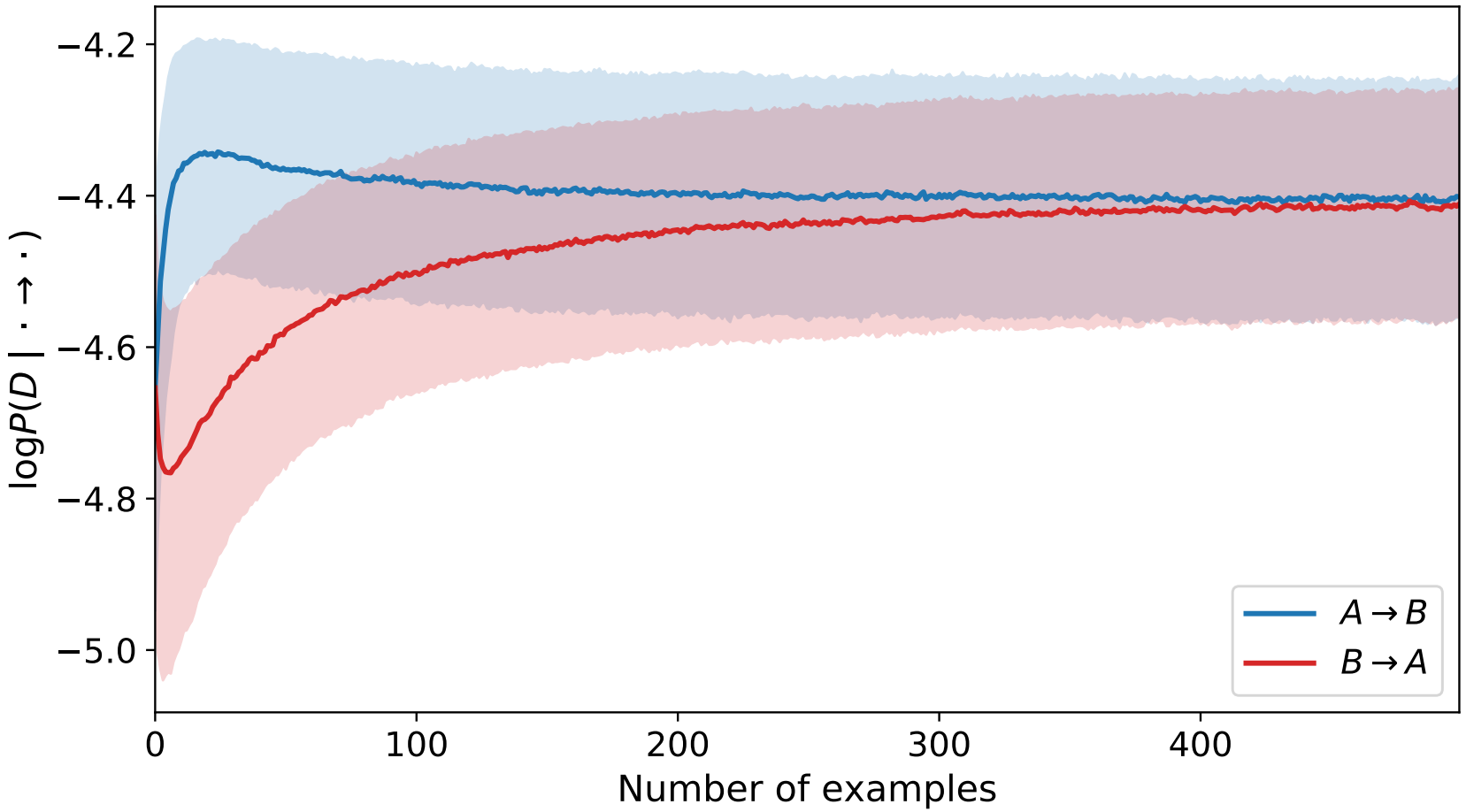
$$\hat{\theta}_i = n_i/n \qquad \hat{\theta}_{j|i} = n_{ij}/n_i$$
$$\hat{\eta}_j = n_j/n \qquad \hat{\eta}_{i|j} = n_{ij}/n_j$$

We can now compute the likelihood for each model:

$$\hat{P}_{A\to B}(A, B) = \hat{\theta}_i \hat{\theta}_{j|i} = n_{ij}/n$$
$$\hat{P}_{B\to A}(A, B) = \hat{\eta}_j \hat{\eta}_{i|j} = n_{ij}/n$$

Which direction can adapt faster? Answer: **The causal direction**

Simulation


−4.2
−4.4
−4.6
−4.8
−5.0
$\log P(D \mid \cdot \rightarrow \cdot)$
0
100
200
300
400
Number of examples
$A \rightarrow B$
$B \rightarrow A$

Proposition

The expected gradient over the transfer distribution of the regret (accumulated negative log-likelihood during the adaptation episode) with respect to the module parameters is zero for the parameters of the modules that (a) were correctly learned in the training phase, and (b) have the correct set of causal parents, corresponding to the ground truth causal graph, if (c) the corresponding ground truth conditional distributions did not change from the training distribution to the transfer distribution.

As a consequence, the effective number of parameters that need to be adapted, when one has the correct causal graph structure, is reduced to those of the mechanisms that actually changed from the training to the transfer distribution.

Proposition

Consider conditional probability modules $P_{\theta_i}(V_i|\mathrm{pa}(i, V, B_i))$ *where* $B_{ij} = 1$ *indicates that* V_j *is among the parents* $\mathrm{pa}(i, V, B_i)$ *of* V_i *in a directed acyclic causal graph. Consider ground truth training distribution* P_1 *and transfer distribution* P_2 *over these variables, and ground truth causal structure* B*. The joint log-likelihood* $\mathcal{L}(V)$ *for a sample* V *with respect to the module parameters* θ *decomposed into module parameters* θ_i *is* $\mathcal{L}(V) = \sum_i \log P_{\theta_i}(V_i|\mathrm{pa}(i, V, B_i))$*. If (a) a model has the correct causal structure* B*, and (b) it been trained perfectly on* P_1*, leading to estimated parameters* θ*, and (c) the ground truth* P_1 *and* P_2 *only differ from each other only for some* $P(V_i|\mathrm{pa}(i, V, B_i))$ *for* $i \in C$*, then* $E_{V \sim P_2}[\frac{\partial \mathcal{L}(V)}{\partial \theta_i}] = 0$ *for* $i \notin C$*.*

Bi-Variate Example

The transfer distribution only changed the true $P(A)$ (the cause)

- For the correct model only $N-1$ parameters need to be re-estimated.
- In the backward model, all $N(N-1)+(N-1)=N^2-1$ parameters must be re-estimated.

More than two parameters

- We won't be able to enumerate all DAGs and pick the best one after observing episodes of adaptation.
- We can parameterize our belief about an exponentially large set of hypotheses by keeping track of the probability for each directed edge of the graph to be present

Formulization

- Modeling edges

$$B_{ij} \sim \mathrm{Bernoulli}(p_{ij}), \quad P(B) = \prod_{ij} P(B_{ij}).$$

- The parents of V_i, given B, as the set of V_j's such that $B_{ij} = 1$:

$$\mathrm{pa}(i, V, B_i) = \{V_j \mid B_{ij} = 1,\ j \neq i\}$$

- The Structural Causal Model:

$$V_i = f_i(\theta_i, B_i, V, N_i)$$

  - N_i is an independent noise source to generate V_i
  - f_i parametrizes the generator (active if $B_{ij} = 1$)

- The conditional likelihood $P_{B_i}(V_i = v_{ti} \mid \mathrm{pa}(i, v_t, B_i))$ measures how well the model that uses the incoming edges B_i for node i performs for example v_t.

$$\mathcal{L}_{B_i} = \prod_t P_{B_i}(V_i = v_{ti} \mid \mathrm{pa}(i, v_t, B_i)). \tag{1}$$

- The overall exponentiated regret for the given graph structure B is

$$\mathcal{L}_B = \prod_i \mathcal{L}_{B_i}$$

- for the generalized multi-variable case

$$\mathcal{R} = -\log E_B[\mathcal{L}_B] \tag{2}$$

Proposition

The overall regret (Equation (2)) rewrites

$$\mathcal{R} = -\sum_i \log \sum_{B_i} P(B_i)\mathcal{L}_{B_i} \tag{3}$$

and if we are willing to consider multiple samples of B in parallel, a biased but asymptotically unbiased (as the number K of these samples $B^{(k)}$ increases to infinity) estimator of the gradient of the overall regret with respect to meta-parameters can be defined:

$$g_{ij} = \frac{\sum_k (\sigma(\gamma_{ij}) - B_{ij}^{(k)})\mathcal{L}_{B_i}^{(k)}}{\sum_k \mathcal{L}_{B_i}^{(k)}} \tag{4}$$

where the $^{(k)}$ index indicates the values obtained for the k-th draw of B.

Recall that $\mathcal{L}_B = \prod_i \mathcal{L}_{B_i}$ so we can rewrite the regress loss as follows:

$$\begin{aligned}
\mathcal{R} &= -\log E_B[\mathcal{L}_B] \\
&= -\log \sum_B P(B)\mathcal{L}_B \\
&= -\log \sum_{B_1} \sum_{B_2} \cdots \sum_{B_M} \prod_i P(B_i)\mathcal{L}_{B_i} \\
&= -\log \prod_i \left(\sum_{B_i} P(B_i)\mathcal{L}_{B_i} \right) \\
&= -\sum_i \log \sum_{B_i} P(B_i)\mathcal{L}_{B_i}
\end{aligned}$$

So the regret gradient on meta-parameters γ_i of node i is

$$\frac{\partial \mathcal{R}}{\partial \gamma_i} = \frac{E_{B_i}[\mathcal{L}_{B_i} \frac{\partial \log P(B_i)}{\partial \gamma_i}]}{E_{B_i}[\mathcal{L}_{B_i}]}$$

Note that with the sigmoidal parametrization of $P(B_{ij})$,

$$\log P(B_{ij}) = B_{ij} \log \mathrm{sigmoid}(\gamma_{ij}) + (1 - B_{ij}) \log(1 - \mathrm{sigmoid}(\gamma_{ij}))$$

as in the cross-entropy loss. Its gradient can similarly be simplified to

$$\begin{aligned} \frac{\partial \log P(B_{ij})}{\partial \gamma_{ij}} &= \frac{B_{ij}}{\mathrm{sigmoid}(\gamma_{ij})} \mathrm{sigmoid}(\gamma_{ij})(1 - \mathrm{sigmoid}(\gamma_{ij})) \\ &\quad - \frac{(1 - B_{ij})}{(1 - \mathrm{sigmoid}(\gamma_{ij}))} \mathrm{sigmoid}(\gamma_{ij})(1 - \mathrm{sigmoid}(\gamma_{ij}))) \\ &= B_{ij} - \mathrm{sigmoid}(\gamma_{ij}) \end{aligned} \tag{5}$$

$$\frac{\partial \mathcal{R}}{\partial \gamma_i} = \frac{E_{B_i}[\mathcal{L}_{B_i} \frac{\partial \log P(B_i)}{\partial \gamma_i}]}{E_{B_i}[\mathcal{L}_{B_i}]}$$

A biased but asymptotically unbiased estimator of $\frac{\partial \mathcal{R}}{\partial \gamma_{ij}}$ is thus obtained by sampling K graphs (over which the means below are run):

$$g_{ij} = \sum_k (\sigma(\gamma_{ij}) - B_{ij}^{(k)}) \frac{\mathcal{L}_{B_i^{(k)}}}{\sum_{k'} \mathcal{L}_{B_i^{(k')}}} \tag{6}$$

where index $^{(k)}$ indicates the k-th draw of B

Representation Learning

- So far, we have assumed that the system has unrestricted access to the true underlying causal variables, A and B.
- This is not always the case!

$$\begin{bmatrix} X \\ Y \end{bmatrix} = R(\theta_{\mathcal{D}}) \begin{bmatrix} A \\ B \end{bmatrix}$$

- If this is the case, our working assumption – that the correct causal graph will be sparsely connected, made of independent components, and affected sparsely by distributional shifts – can not be expected to hold true in general in the space of observed variables.

Learn the encoder parameters as well!

Appendix C. Pseudo-Code

```
Draw initial meta-parameters of learner
Draw a training set from training distr.
Set causal structure to include all edges
Initialize learner parameters for this model
Pre-train the learner's parameters on the training set
Repeat J times
    Draw a transfer distr.
    Draw causal structure(s) according to meta-parameters
    Repeat T times
        Sample minibatch from transfer distribution
        Accumulate online log-likelihood of minibatch
        Update the model parameters accordingly

    Compute the meta-parameters gradient estimator
    Update the meta-parameters by SGD
    Optionally reset parameters to pre-training value
```

Future Works!

This work is only a first step in the direction of optimizing causal structure based on the speed of adaptation to modified distributions.

- This paper has only tested this on artificial data! Can we think of an application?
- The representation learning has only been experimentally tested for a very simple rotation matrix!
- Convergence rates for SGD.
- Biased estimator for the gradient!
- On the experimental side, many settings other than those studied here should be considered, with different kinds of parametrizations, richer and larger causal graphs, different kinds of optimization procedures, etc.